

Name: _____ Student ID: _____

Midterm 1 (Max. score = 80 points)

Math 589B, Spring 2024

Q 1. (10 points) Given is the following Newton's divided difference table of the function $f(x)$:

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}, x_{i+4}]$
0	0	0.7956	0.1312	0.2379	0.1792
0.25	0.1989	0.8612	0.3096	0.4171	
0.50	0.4142	1.016	0.6224		
0.75	0.6682	1.3272			
1.00	1				

Use the 4th-degree Newton polynomial constructed from the table to approximate $f(0.1)$. Make sure to give the answer with the correct number of significant digits, assuming that the values in the table are correctly rounded to 4 significant digits, i.e., have relative error $\leq 0.5 \times 10^{-4}$ (1/2 ulp guarantee).

THE SCRATCH AREA

0.07832, rounded to **five digits** after the decimal point

THE ANSWER

- Q 2.** In numerical integration, the Midpoint Rule (for a single subinterval) approximates the definite integral of a function f over $[a, b]$ by evaluating f at the midpoint $m = \frac{a+b}{2}$:

$$\int_a^b f(x) dx \approx (b - a) f\left(\frac{a + b}{2}\right).$$

- Q 2.1.** (10 points) Write down (in your own words) the statement of the Midpoint Rule for approximating an integral of a continuous function f over the interval $[a, b]$.

THE PROOF

QED

- Q 2.2.** (10 points) Using ideas from how the error was derived for the Trapezoidal Rule (which we discussed in class), derive an error expression of the form

$$\text{Error} = -\frac{(b-a)^3}{24} f''(\xi)$$

for some $\xi \in (a, b)$. (Hint: consider the second derivative of f , apply an appropriate form of the Remainder Theorem or Taylor expansion, and carefully compare the exact integral to the midpoint approximation.)

THE SCRATCH AREA

$$\text{Error} = \frac{f''(\xi)}{24} (b-a)^3$$

THE ANSWER

- Q 2.3.** (10 points) Give a detailed proof that shows why there must exist such a point ξ in (a, b) satisfying the error bound. (You may use the Mean Value Theorem for integrals, Rolle's Theorem or other standard theorems covered in class and/or calculus.)

THE PROOF

QED

- Q 3.** Consider the Legendre polynomial of degree 3, $P_3(x)$. By definition, Legendre polynomials can be obtained from the *Rodrigues formula*:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n].$$

- Q 3.1.** (10 points) Use the Rodrigues formula with $n = 3$ to derive an explicit expression for $P_3(x)$.

THE SCRATCH AREA

$$P_3(x) = \frac{5x^3 - 3x}{2}$$

THE ANSWER

Q 3.2. (10 points) Find the roots of $P_3(x)$. These roots will be the *nodes* $x_1 < x_2 < x_3$ for the 3-point Gauss–Legendre quadrature rule on the interval $[-1, 1]$.

THE SCRATCH AREA

THE ANSWER

Q 3.3. (10 points) The standard formula for the *weights* w_i in Gauss–Legendre quadrature is

$$w_i = \frac{2}{(1 - x_i^2) [P'_n(x_i)]^2},$$

where $P'_n(x)$ is the derivative of the Legendre polynomial of degree n . Compute the weights w_1, w_2, w_3 for $n = 3$, using the roots x_i from part (b).

THE SCRATCH AREA

THE ANSWER

- Q 3.4.** (10 points) List your final answers for the *nodes* and *weights* $\{(x_1, w_1), (x_2, w_2), (x_3, w_3)\}$ in ascending order of the x_i . Hence, your completed 3-point Gauss–Legendre rule will have the form

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^3 w_i f(x_i).$$

THE SCRATCH AREA

$$\begin{aligned} x_1 &= -\sqrt{\frac{3}{5}}, & w_1 &= \frac{5}{9}, \\ x_2 &= 0, & w_2 &= \frac{8}{9}, \\ x_3 &= \sqrt{\frac{3}{5}}, & w_3 &= \frac{5}{9}. \end{aligned}$$

THE ANSWER

CONTINUATION OF PROBLEM:

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